Chronic Kidney Disease: Creatinine, eGFR, and Medication Safety

This is a synthetic, simplified reference document created for a portfolio software project. It is not real clinical guidance and must not be used for patient care.

Estimated glomerular filtration rate (eGFR) and serum creatinine are the primary markers used to stage and monitor chronic kidney disease. A reference eGFR of 90-120 mL/min/1.73m2 is considered normal; sustained values below 60 for three months or more are diagnostic of chronic kidney disease, and values below 15 represent kidney failure requiring urgent nephrology involvement.

Creatinine values should be interpreted alongside the patient's baseline, since a 'normal range' value can still represent acute kidney injury in a patient whose baseline was previously lower. A critical creatinine result above 4.0 mg/dL should prompt urgent evaluation regardless of the patient's chronic baseline.

Metformin is renally cleared, and its use in patients with reduced eGFR requires dose adjustment or discontinuation below certain thresholds due to the risk of lactic acidosis. This risk is further increased by exposure to iodinated contrast media, which can transiently worsen renal function; metformin should be held before and for 48 hours after contrast administration in at risk patients.

Patients with chronic kidney disease on ACE inhibitors or potassium-sparing diuretics require closer potassium and creatinine monitoring than patients with normal renal function, since impaired excretion increases the risk of both hyperkalemia and further renal decline.